



TRIOS CYBER

LABORATORY PRACTICE 11

Log Filtering and Pattern Detection

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1. Sample Log File Creation

A small authentication-style sample log file named `sample_auth.log` was created on the Ubuntu laboratory endpoint using `nano`. Seven records were entered with timestamps, usernames, source IP addresses, authentication results, and message fields. The dataset intentionally included repeated values so that filtering and pattern detection could be demonstrated with command-line tools.

```
matthew@matthew-VirtualBox:~$ nano sample_auth.log
```

```
GNU nano 4.8 sample_auth.log Modificado
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_username
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
```

The completed file was displayed with `cat` to verify that all seven records had been saved correctly. The dataset contained the users `matthew`, `admin`, and `test`; source addresses `10.10.10.20`, `10.10.10.30`, and `10.10.10.40`; six `FAILED` results; and one `SUCCESS` result.

```
matthew@matthew-VirtualBox:~$ cat sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:15 user=matthew src=10.10.10.20 result=SUCCESS message=Accepted_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_username
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
```

2. Failed Authentication Filtering with grep

The sample file was filtered with `grep` using the pattern `result=FAILED`. This isolated the six unsuccessful records and excluded the single successful authentication entry, providing a focused dataset for repeated-pattern analysis.

```
matthew@matthew-VirtualBox:~$ grep "result=FAILED" sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_username
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
```

3. Failure Message Filtering

A second grep search was performed for the Failed_password message. The output showed five records containing the same password-failure message, confirming that this text pattern occurred repeatedly in the sample data.

```
matthew@matthew-VirtualBox:~$ grep "result=FAILED" sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
matthew@matthew-VirtualBox:~$ grep "Failed_password" sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
```

4. Repeated Username Detection with awk and sort

The username field was extracted with awk and ordered with sort. A counting expression then summarized the frequency of each username. The results showed matthew four times, admin twice, and test once, making matthew the most frequently observed username in the dataset.

```
matthew@matthew-VirtualBox:~$ awk '{print $2}' sample_auth.log | sort
user=admin
user=admin
user=matthew
user=matthew
user=matthew
user=matthew
user=test
matthew@matthew-VirtualBox:~$ awk '{count[$2]++} END {for (user in count) print count[user], user}' sample_auth.log | sort -nr
4 user=matthew
2 user=admin
1 user=test
```

5. Repeated Source IP Detection

The same counting approach was applied to the source-address field. The analysis identified 10.10.10.20 four times, 10.10.10.30 twice, and 10.10.10.40 once. This demonstrated how repeated source systems can be identified quickly from structured log data.

```
matthew@matthew-VirtualBox:~$ awk '{count[$3]++} END {for (ip in count) print count[ip], ip}' sample_auth.log | sort -nr
4 src=10.10.10.20
2 src=10.10.10.30
1 src=10.10.10.40
```

6. Repeated Failure Message Detection

Failed records were filtered first and the message field was then counted with awk and sorted by frequency. Failed_password appeared five times, while Invalid_user appeared once. The result highlighted the dominant failure pattern in the sample log.

```
matthew@matthew-VirtualBox:~$ grep "result=FAILED" sample_auth.log
2026-09-12T13:00:01 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:05 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
2026-09-12T13:00:10 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:20 user=admin src=10.10.10.30 result=FAILED message=Failed_password
2026-09-12T13:00:25 user=test src=10.10.10.40 result=FAILED message=Invalid_user
2026-09-12T13:00:30 user=matthew src=10.10.10.20 result=FAILED message=Failed_password
matthew@matthew-VirtualBox:~$ grep "result=FAILED" sample_auth.log | awk '{count[$5]++} END {for (msg in count) print count[msg], msg}' | sort -nr
5 message=Failed_password
1 message=Invalid_user
```

7. Combined Pattern Filter

A combined pipeline used grep, awk, and sort to display only failed records while retaining the username, source IP, and message fields. This compact output made it possible to compare the repeated indicators in a single view without reviewing unrelated fields.

```
matthew@matthew-VirtualBox:~$ grep "result=FAILED" sample_auth.log | awk '{print $2, $3, $5}' | sort
user=admin src=10.10.10.30 message=Failed_password
user=admin src=10.10.10.30 message=Failed_password
user=matthew src=10.10.10.20 message=Failed_password
user=matthew src=10.10.10.20 message=Failed_password
user=matthew src=10.10.10.20 message=Failed_password
user=test src=10.10.10.40 message=Invalid_user
```

8. Pattern Detection Summary

The filtered results were consolidated to identify the most important repeated values observed in the laboratory sample.

Pattern Type	Observed Value	Count	Interpretation
Username	matthew	4	Most frequent username
Username	admin	2	Repeated account value
Source IP	10.10.10.20	4	Most frequent source address
Source IP	10.10.10.30	2	Repeated source address
Failure message	Failed_password	5	Dominant failure message
Failure message	Invalid_user	1	Single alternative failure message

9. Log Filtering Workflow Review

The activity demonstrated a simple command-line analysis workflow: grep was used to isolate relevant records, awk extracted and counted structured fields, and sort organized repeated values so that dominant patterns could be recognized quickly. This approach can support initial SOC log review by reducing large datasets to the usernames, source addresses, and failure messages that occur most frequently.

10. Conclusions

The log filtering and pattern-detection activity was successfully completed on the Ubuntu Linux laboratory endpoint. A seven-record sample log was created and analyzed with grep, awk, and sort. Six failed records were isolated, repeated usernames and source IP addresses were counted, and Failed_password was identified as the dominant failure message with five occurrences. The exercise demonstrated how basic Linux text-processing tools can rapidly identify recurring indicators that may warrant closer investigation in larger security logs.